

HINTS AND SOLUTIONS

Q.1 B

Q.2 C

Sol. Constancy of flux implies that $\frac{E}{R} \cdot L_1 = i(L_1 + L_2)$

$$\text{i.e., } i = \frac{E L_1}{R(L_1 + L_2)}$$

Q.3 D

Sol. Potential drop is only across wire AB

$$\phi = B_0 \sin(\omega t) \cdot \left(\frac{\pi r^2}{6} \right)$$

$$\varepsilon = \left| \frac{-d\phi}{dt} \right| = \frac{B_0 \omega \pi r^2}{6}$$

Q.4 C

Sol. $\varepsilon = \frac{d\phi}{dt}$, where $\phi = B\pi (R_0 + t)^2$.

Q.5 D

Sol. Effective emf $(n-2)\varepsilon$

$$i = \frac{(n-2)\varepsilon - Bv\ell}{r}$$

Q.6 C

Sol. Friction force acting on the bottom end of the ring will be zero.

Linear impulse = change in linear momentum

$$\text{or, } F \times t = m \{V_c - 0\}$$

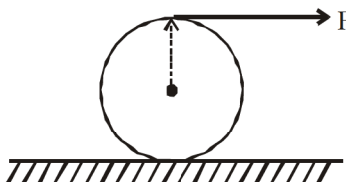
$$\text{or, } V_c = \frac{F \times t}{m}$$

$$\varepsilon_{\text{ind}} = B \times 2R \times V_c$$

$$= \frac{2BR \times F \times t}{m}$$

$$= \frac{2FBRt}{m}$$

Because, there will be no any induced emf due to angular velocity w.r.t. to COM of the ring



- Q.8 B
Q.9 A
Q.10 C
Q.11 A
Q.12 B
Q.13 C
Q.14 C
Q.15 C

Q.16 [Ans. (A) Q,S (B) P,R (C) P,R]

Q.17 Ans. 3.14, 3.3, 200/3.3 A, $\tan^{-1}\pi$

Sol. The reactance of the circuit = $X_L = L\omega = 2\pi fL$
 $\therefore X_L = 0.01 \times 2\pi \times 50 = \pi = 3.14\Omega$

The impedance of the circuit = $\sqrt{X_L^2 + R^2}$

$$\therefore Z = \sqrt{(3.14)^2 + 1^2} = \sqrt{10.86} = 3.3\Omega$$

The phase difference (lag) is
 $\tan\phi = L\omega / R = \pi/1 = \pi$

Q.18 [Ans. 8×10^{-8} volts]

Sol. $\phi = \frac{\mu_0 i}{2\pi R} \times N \times A$

$$r = \frac{\mu_0 NA}{2\pi R} \times k = \frac{4\pi^2 \times 10^{-7} \times 10^{-4} \times 8^2 \times 2000}{2\pi \times 4} = 8 \times 10^{-8} \text{ volts}$$

Q.19 Ans. 0.08 H, 17.28 W

Sol. When connected to d.c. source

$$R = \frac{V}{I} = \frac{12}{4} = 3\Omega \quad \dots(1)$$

When connected to a.c. source

$$Z = \frac{V_{\text{rms}}}{I_{\text{rms}}} = \frac{12}{2.4} = 5\Omega \quad \dots(2)$$

$$Z = \sqrt{R^2 + (\omega L)^2} \quad \dots(3)$$

From (1), (2) and (3)

$$L = 0.08 \text{ H}$$

with condenser

$$\begin{aligned} Z' &= \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \\ &= \sqrt{3^2 + \left(50 \times 0.08 - \frac{1}{50 \times 2500 \times 10^{-6}}\right)^2} \\ &= 5\Omega \end{aligned}$$

$$\cos \phi = \frac{R}{Z'} = 0.6$$

$$\begin{aligned} \therefore P &= V_{\text{rms}} I_{\text{rms}} \cos \phi \\ &= 12 \times 2.4 \times 0.6 = 17.28 \text{ W} \end{aligned}$$



Q.20 Ans. (i) 1 H, (ii) 434 μF

Sol. Current lags behind potential difference means circuit contains resistance and inductance.

$$\therefore P = V_{\text{rms}} i_{\text{rms}} \cos \phi = \frac{V_{\text{rms}}^2}{Z} \cos \phi$$

$$\Rightarrow Z = \frac{V_{\text{rms}}^2 \cos \phi}{P} = \frac{(200)^2 \times 0.8}{400} = 80 \Omega$$

$$\therefore R = Z \cos \phi = 80 \times 0.8 = 64 \Omega$$

$$(a) \omega L = \sqrt{Z^2 - R^2} = \sqrt{(80)^2 - (64)^2} = \sqrt{2304} = 48$$

$$L = \frac{48}{2\pi f} = \frac{48}{2\pi \times \frac{24}{\pi}} = 1 \text{ H}$$

$$(b) \frac{1}{\omega C} = \omega L$$

$$\omega C = \frac{1}{\sqrt{Z^2 - R^2}} = \frac{1}{\sqrt{(80)^2 - (64)^2}} = \frac{1}{\sqrt{2304}} = \frac{1}{48}$$

$$\Rightarrow C = 434 \mu\text{F}$$

Q.21 Ans. $M = \frac{\mu_0 a}{2\pi} \ln \frac{(a+\ell)^2}{\ell(2a+\ell)}$, $\varepsilon = -M I_0$, Heat produced $= \frac{\varepsilon^2}{R} t$

Sol. Consider a strip at a distance x from the wire of thickness dx . Magnetic flux associated with this strip

$$\Delta \phi = B(x) a dx$$

$$= \frac{\mu_0 I a}{2\pi x} dx$$

$$\Rightarrow \phi = \frac{\mu_0 I a}{2\pi} \left[\int_{\ell}^{a+\ell} \frac{dx}{x} - \int_{a+\ell}^{2a+\ell} \frac{dx}{x} \right] = \frac{\mu_0 I a}{2\pi} \ln \frac{(a+\ell)^2}{\ell(2a+\ell)}$$

$$M = \frac{\mu_0 a}{2\pi} \ln \frac{(a+\ell)^2}{\ell(2a+\ell)}$$

$$(b) \therefore \varepsilon = -M \frac{dI}{dt}$$

$$\Rightarrow e = -M I_0$$

$$\text{Heat produced} = \frac{\varepsilon^2}{R} t$$

